

Drug d

Drug profile $\log C_{free}$ $\text{pAC}_{50} \times 194$ $\text{logratio} \times 194$

389 input features

 $\text{logratio}_i = \log_{10} C_{max, free} + \text{pAC}_{50, i} - 6$

OT-ProfileNet + exposure models

Drug encoder
389–512–256–128 h_d **Pair representation**
 $|h_d - h_a|$ **Pair MLP**
64–32–16–8**SOC head**
 $w_a^T h + b_a$
sigmoid**P(ADR_a)**
18 SOC_s

SOC a

SOC index
1–18Trainable
embedding 512**ADR encoder**
512–256–128 h_a **Training objective**

BCE: real + soft pseudo-dose labels

Monotone: $0.5 \cdot \text{ReLU}(p_{low} - p_{high})$

— same drug/SOC, low → high dose

CT-ADE fine-tuning
SOC heads only**5-fold ensemble**